Solutions of the last two problems in the training lesson on 9/8

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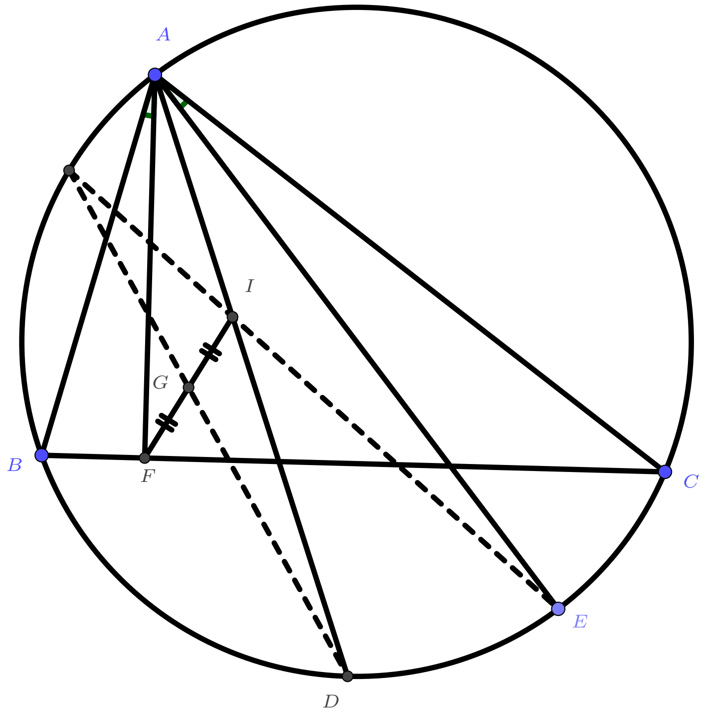
Here are the idea flows for the last two hard problems. (Okok it's a belated piece of notes but it finally appears right *smile*)

Problem (IMO 2010 Problem 2)**.** Given a triangle ABC, with I as its incenter and Γ as its circumcircle, AI intersects Γ again at D. Let E be a point on the arc BDC, and F a point on the segment BC, such that $\angle BAF = \angle CAE < \frac{1}{2}\angle BAC$. If G is the midpoint of IF, prove that the meeting point of the lines EI and DG lies on Γ.

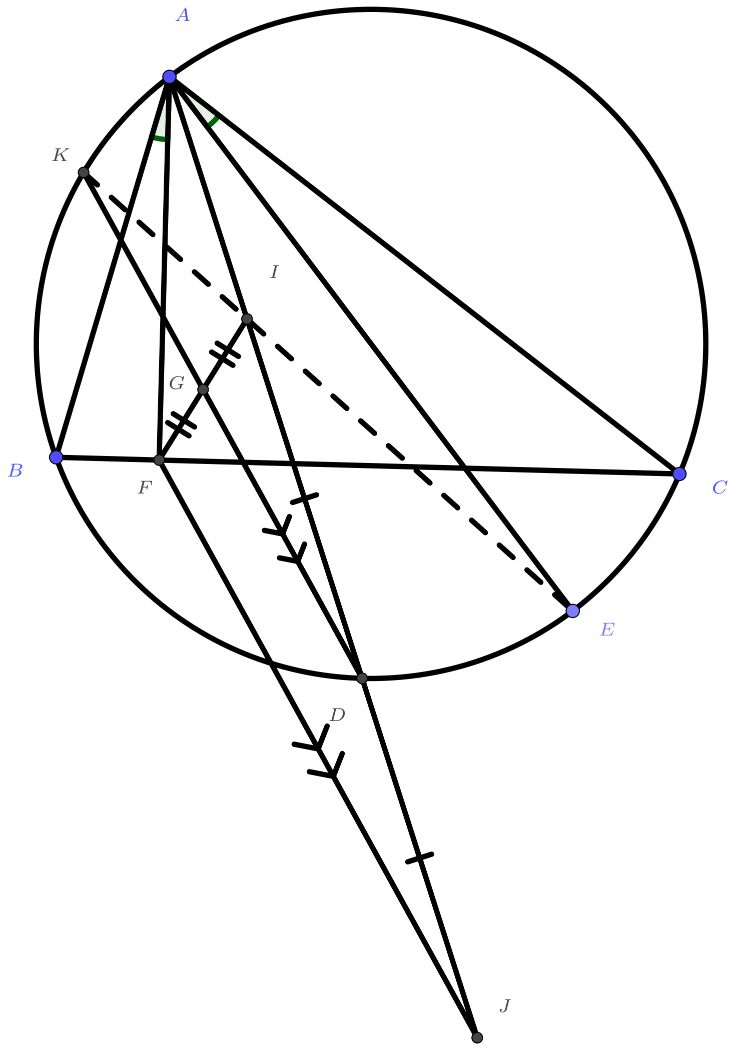
Solution. In the first place, to tackle with a hard problem, one shall draw an accurate figure:



How to prove the intersection of two lines lying on a circle? To prove concurrence, one often uses Ceva's theorem, yet it obviously doesn't work here. So what to do? Perhaps simply by angles? The question setting also got a pair of equal angles, what could it give even? How do we use the midpoint condition? There must be something else constructed for the sake of using the midpoint. But what to construct? Well, at least the ways to use the midpoint is constructive and limited so you may try them all. In fact how to utilise the midpoint sort of hints the solutions.

My Solution

We would like to treat $G\ddot{\smile}$ as a midline of $\triangle IF\ddot{\frown}$ for some $\ddot{\smile}$ and $\ddot{\frown}$ so that we can use the midpoint theorem. For the whole thing to be useful, one usually construct either an extra $\ddot{\smile}$ or $\ddot{\frown}$ but not both. After trial and error, one finds that this construction makes the best sense among:

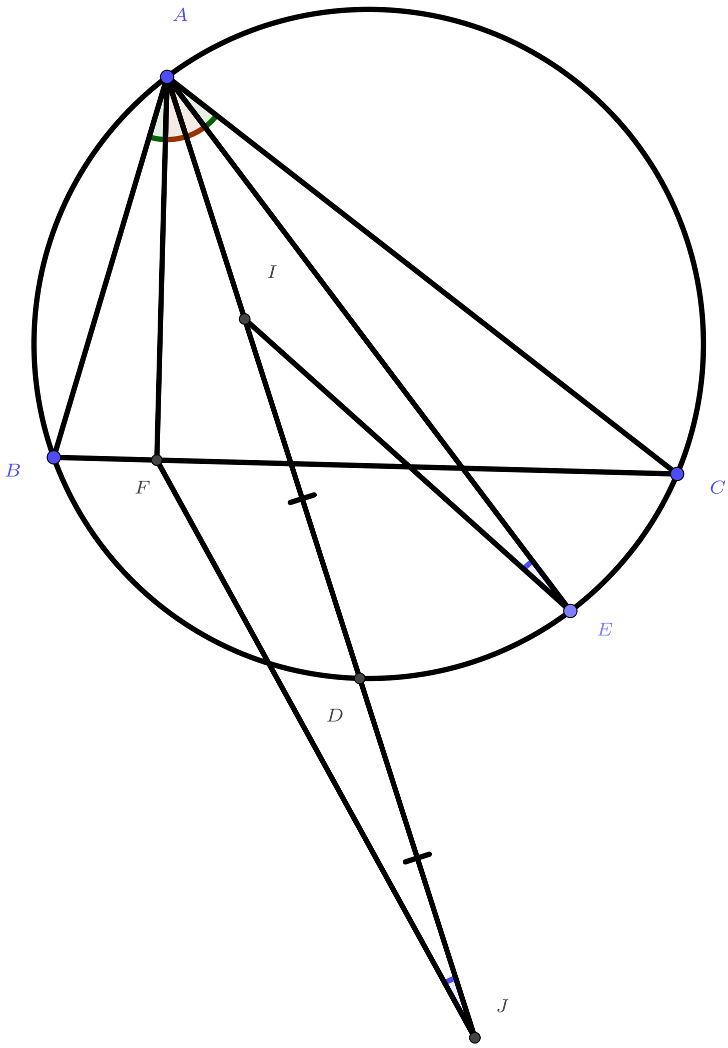


Why? In the first place, GD already existed and is one of the lines that we concern the most, so trying to related GD with another thing would be a reasonable move. Also, if one recalled some extra properties related to the excentre, **one would realise J is the excentre of $\triangle ABC$!** One shall recall the figure that D is actually the centre of the circle $BICJ$ with diameter IJ ($\angle IBJ = \angle ICJ = 90°$). Finally, after we obtain the parallel property by length ratio (in this case it's 1:1, still about the side lengths), the information we shall extract from it shall be angles. Now it's not difficult to spot our final aim: if DG meets Γ again at K, then

$$\begin{aligned} EI \cap DG \in \Gamma \iff K, I, E \text{ collinear} &\iff \angle AEK = \angle AEI \\ \iff \angle ADK = \angle AEI &\iff \angle AJF = \angle AEI. \end{aligned}$$

Tada! Notice **how the points G and K completely disappear in our problem statement.** Now you can totally erase these two points for a simplier picture.

Now what next? Notice that we haven't used the angle relation yet. Since AI is the angle bisector of $\angle BAC$, the angle relation means that lines AF and AE are symmetric with respect to line AI. Recalling our new aim $\angle AJF = \angle AEI$, it's perhaps clear that what we should approach:



Simple angle chasing yields $\angle FAJ = \angle IAE$. Hence to prove that $\angle AJF = \angle AEI$, it is equivalent to show $\triangle AFJ \sim \triangle AIE$. Since we want to prove angle relation out from the pair of similar triangles, one should prove the similar triangles by lengths ratios. As $\angle FAJ = \angle IAE$ is already given, we only need to show

$$\frac{AF}{AJ} = \frac{AI}{AE}, \text{ or } AF \cdot AE = AI \cdot AJ.$$

Now $AI \cdot AJ$ is something that could be computed by the original triangle, i.e. a, b, c, so even if you don't know how to compute it, at least you know it's something "computable". So we should try to find how to express $AF \cdot AE$ in terms of the original triangle instead — we should show that it's a constant regardless of the position of E! (Recall that E is constructed in an arbitrary way.)

Notice that E moves freely around the arc, and when it coincides with C, F coincides with B, and one anticipates that the product $AF \cdot AE$ is kept constant. Hence one expects that $AF \cdot AE = AB \cdot AC$. Now given that we want to prove $AF \cdot AE = AB \cdot AC = AI \cdot AJ$, it's not difficult to observe $\triangle ABF \sim \triangle AEC$ and $\triangle ABJ \sim \triangle AIC$ (by angle relations, of course, since we shall obtain information of lengths from so), and we are done.

Other Ideas and Remarks

Of course, if one knows something about isogonal lines, or even $\sqrt{bc}$-inversion (Insanity alert! It's one of the favourite tools of a former trainer, yet it's too difficult for most trainees $\ddot\frown$), then one would be much easier to observe those above, especially those similar triangles. What else could one approach this problem?

If one draws the diagram really very accurately, then it may be observed that if $P = DG \cap AF$, then PI do looks like being parallel to BC. From this point we can form a few ideas attempting:

One other way to use the midpoint condition is to form parallelograms. If we have a midpoint G of IF together with IP being parallel to BC, then there should be a parallelogram appearing — denote $Q = DG \cap BC$, then $PIQF$ would be a parallelogram! In fact one can form another solution by letting $EI \cap \Gamma = K$, $KD \cap AF = P$, $KD \cap IF = G'$, $KD \cap BC = Q$, then trying to prove $PIFQ$ being a parallogram could prove $G = G'$.

(Insanity alert again) If one knows projective geometry, since you have midpoints and parallel, it may hint you about some hidden harmonic relations in the figure too. (One need to be very familiar with projective geometry before actively using it to solve problems though. If one's not even familiar with usual geometry tools, then attempting to use such an extra stuff may risk in having no actual progress at all during competitions.)

This problem can also be achieved by trigonometry and length chasing. Of course, observing similar triangles $\triangle BAF$ and $\triangle EAC$ still helps much, since that's the best way to make use of $\angle BAF = \angle CAE$ anyway.

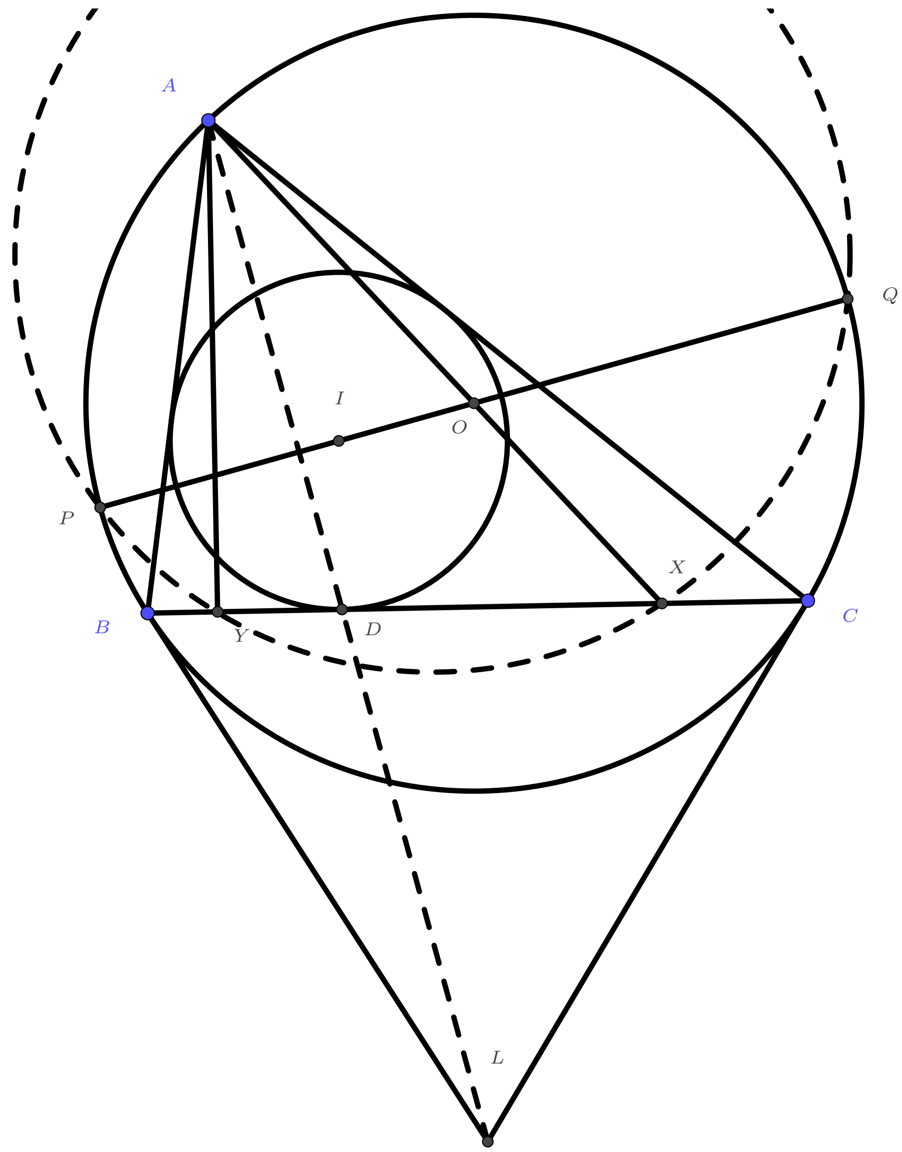
If one really wants to use coordinate geometry or complex geometry, do note that some pure geometry observations are still very helpful, if not necessary, in order to help with the co-geom progress. Setting a nice origin and direction often helps a lot too. Say, for this problem, one may set the only circle as the unitary circle in complex plane and let the complex numbers deal with the angles. If not, then angles is a huge concern, and one may set A as the origin together with AI being an axis in order to make some better interpretations of the angles.

As a last note, it's G4 in the shortlist, which is often out of reach by some Hong Kong team members simply because it's a geometry problem. Obviously we do wish students solving geometry problems of this level, but even if not, we hope that one can still have some subtle hints of how geometry problems could be solved in general.

As I've said, the following problem is put here only for those who are bored in my lesson. It's often rumoured that the CMO (China Mathematical Olympiad) is more difficult than IMO, but part the difficulty actually lies under the brute force nature of mainland contests. Hong Kong students are often not familiar with expanding everything, exhausting every case, dealing with a lot of summations etc., which often appears in mainland contests. The problems are often a mixture of idea and expansion. If one have a sense of what is "computable" and what is not, then it may be rather helpful in mainland contests: find some useful ideas around until everything is "computable". Anyways let's have a look in this problem:

Problem (CMO 2016 Q2)**.** In acute scalene triangle ABC, let $\odot O$ be its circumcircle, $\odot I$ be its incircle. Tangents at B, C to $\odot O$ meet at L, $\odot I$ touches BC at D. AY is perpendicular to BC at Y, AO meets BC at X, and OI meets $\odot O$ at P, Q. Prove that P, Q, X, Y are concyclic if and only if A, D, L are collinear.

Solution. We draw the diagram first (even if the figure is given in china contests, it's often not an accurate one):

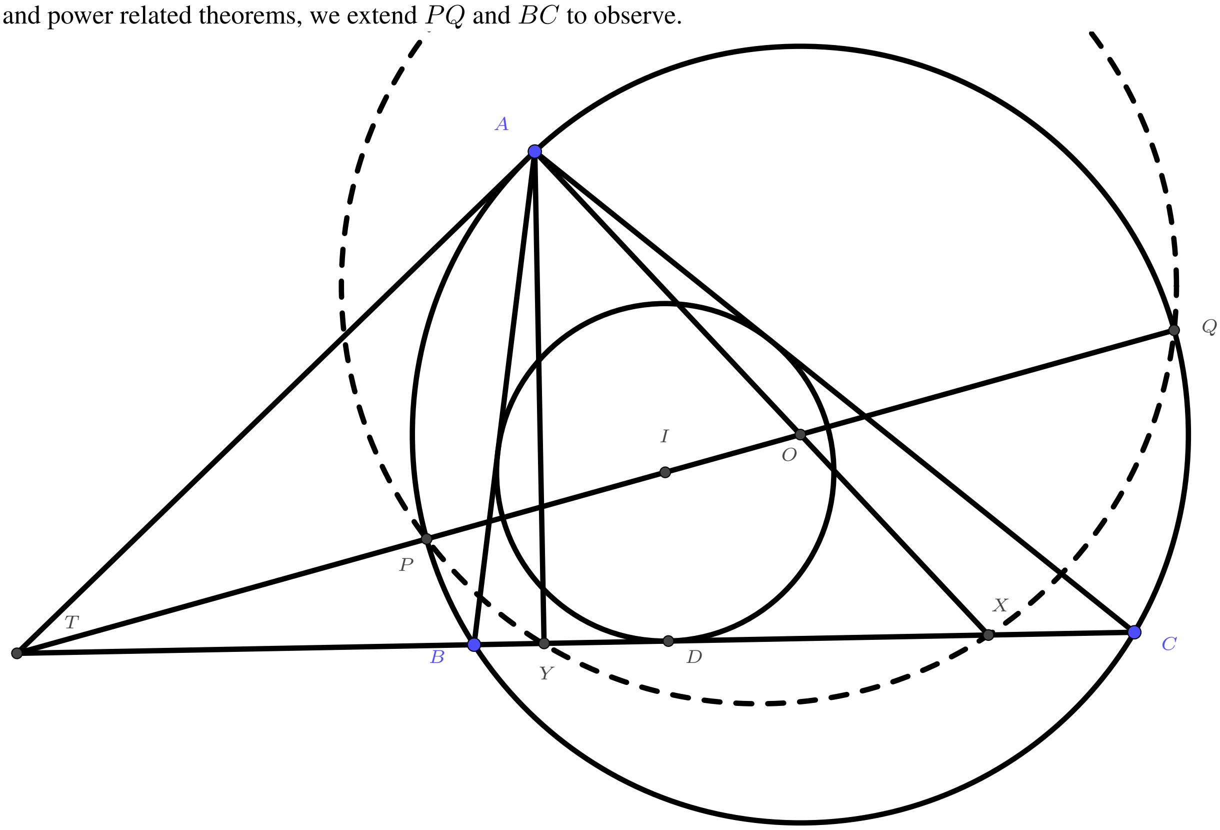


My Solution

Notice that the relation between points D, L and the points P, Q, X, Y are rather weak. One may expect that there exist some intermediate equivalent conditions for the two statements "P, Q, X, Y are concyclic" and "A, D, L are collinear". So perhaps we shall separate P, Q, X, Y and D, L into two figures for clearer observations.

The collinear condition is "easier" to interpret: we 'just' need to recognise AD and AL cutting the angle $\angle BAC$ equally, and obtain some information about $\triangle ABC$. Of course it may end up some rather uncomfortable conditions of a, b, c or even sin and cos of A, B, C, but at least it's computable — it's just technical. On the other hand, the concyclic condition is more weird, since the points PQ seem to be rather irrelevant to other points in the figure, especially X and Y. So what to do?

Compared to P, Q lying on OI, it seems that them lying on $\odot O$ looks even more promising, because PQ will be the **radical axis** of two circles if the concyclic condition holds! Now, to efficiently use the radical axis and power related theorems, we extend PQ and BC to observe.



Notice that P, Q, B, C are always concyclic. Hence if we define $T = BC \cap OI$ (the case where $OI//BC$ is easy to throw away), we must have $TP \cdot TQ = TB \cdot TC$. Now we finally got our first step:

$$P, Q, X, Y \text{ concyclic } \iff TX \cdot TY = TP \cdot TQ \iff TX \cdot TY = TB \cdot TC.$$

Notice how **the points P and Q disappear from the problem statement from now on**. We don't need them any more! Now we need to think of how to interpret $TX \cdot TY = TB \cdot TC$. It seems that the roles of X and Y in the problem are symmetric even if their construction aren't. After all, we don't know much about X besides how AX divides the angle $\angle BAC$. So we either start to impose some brute force or we shall angle chase a while first. Then one shall observe that $\angle BAY = \angle CAX$. (Or perhaps this should already be a fact in mind that $\angle BAH = \angle CAO$!) So they do have some sort of symmetry about angle A.

Since we have the equation that looks like power, we want to interpret it in a power way too. Let ω be the circumcircle of AXY. Then if ω and $\odot O$ meet at A and A', then T would have same power to ω and $\odot O$ iff T lies on AA'. However, upon some sensitive observations on the angle relations, one realise that ω and $\odot O$ shall be tangent to each other! In fact, if S is on BC such that SA is tangent to $\odot O$, then by angle chasing, SA is tangent to ω too, hence it's exactly the radical axis of $\odot O$ and ω. Therefore $SB \cdot SC^2 = SA^2 = SX \cdot SY$, and we expect T to coincide with S. Now we can write

$$TX \cdot TY = TB \cdot TC \iff T \text{ lies on the radical axis of } \odot O \text{ and } \omega \iff TA \text{ is tangent to } \odot O.$$

Notice how **the points X and Y disappear here.** Recall T is the intersection of BC and OI. Then we have totally erased P, Q, X, Y from the problem. Our desired problem statement has become:

Prove that BC, OI and the tangent at A w.r.t. $\odot O$ are concurrent iff A, D, L are collinear.

This is a simplier form of the former condition, since we have transformed the relation of rather lesser known points to that of more known points. The next step is to finally get rid of the latter condition.

For the point L, we only know that $BL = CL$, together with some related angles. Since we're expecting to find the angles $\angle BAL$ and $\angle CAL$, it's then natural to consider sine law. (Since it's mainland competitions we expect it to be sort of brute force! In fact one would think of sine law when there're some equal sides for two triangles, and another pair of sides/angles with known info.) Using the usual notations of $abcABCsRr$, we have

$$\frac{BL}{\sin\angle BAL} = \frac{AL}{\sin ABL} = \frac{AL}{\sin(A+B)} = \frac{AL}{\sin C}$$

and $\dfrac{CL}{\sin\angle CAL} = \dfrac{AL}{\sin B}$, so dividing these two equations yields

$$\frac{\sin\angle BAL}{\sin\angle CAL} = \frac{\sin C}{\sin B} = \frac{c}{b}.$$

On the other hand, line AD divides segment BC into $(s-b) : (s-c)$, hence we know that everything here can be computed. By considering the area we have

$$\frac{s-b}{s-c} = \frac{[\triangle ABD]}{[\triangle ACD]} = \frac{\frac{1}{2}\cdot AB\cdot AD\cdot\sin\angle BAD}{\frac{1}{2}\cdot AC\cdot AD\cdot\sin\angle CAD} = \frac{c\sin\angle BAD}{b\sin\angle CAD},$$

so we have $\dfrac{\sin\angle BAD}{\sin\angle CAD} = \dfrac{b(s-b)}{c(s-c)}$. Therefore, we obtain the equivalent condition for the collinearity:

$$A, D, L \text{ collinear} \iff \frac{c}{b} = \frac{b(s-b)}{c(s-c)} \iff b^2(s-b) = c^2(s-c).$$

At least we achieved to discard the points D and L. Moreover, one observe that this equation must be true if $b = c$ due to symmetry, but since the triangle is scalene we shall throw this case away. Hence one knows that there's a factor $b - c$ which can be thrown away in the equation. In fact we have

$$b^2(s-b) = c^2(s-c) \iff s(b^2-c^2) = b^3 - c^3$$

$$\iff s(b+c) = b^2 + bc + c^2 \iff (b+c)(a+b+c) = 2b^2 + 2bc + 2c^2 \iff a(b+c) = b^2 + c^2.$$

Not too bad as a length relation, huh? Now it remains to prove that BC, OI and the tangent at A w.r.t. $\odot O$ are concurrent iff $a(b+c) = b^2 + c^2$. Let M be the midpoint of BC, then T lying on OI can be expressed into $\dfrac{ID}{TD} = \dfrac{OM}{TM}$, so one can compute the ratio of the location of T on BC. Recall that if S is the intersection of tangent at A w.r.t. $\odot O$ and BC, then $\triangle SAB \sim \triangle SCA$ gives $\dfrac{SA}{SC} = \dfrac{SB}{SA} = \dfrac{AB}{CA} \Rightarrow \dfrac{SB}{SC} = \left(\dfrac{AB}{CA}\right)^2 = \dfrac{c^2}{b^2}$, so that when T coincides with S it must satisfy this ratio too. Therefore by comparing the location of T on BC we can arrive the same condition, and I omit the rather technical computation here for brevity.

Other Ideas and Remarks

The key is to interpret the concyclicity of $PQXY$ in terms of T, which is a crucial step in every solution of this problem. After constructing T and erasing P, Q, theoretically one can already start expanding everything. It is about how ”computable” you think the situation is. In fact most solutions of geometry problems of mainland contests (CMO, CGMO and CWMI) do look ugly.

If one is sensitive enough for projective geometry, one may come up with a better interpretation of the collinearity of A, D, L, and even have a much easier solution (in fact there’re a lot of neat solutions of this problems on AoPS, with most of them using projective geometry though). However, mainland students are not really well trained in projective geometry, and it’s never the official solution anyway. Instead, the above piece of algebraic mess may seem to be an alternate solution, yet it’s more well trained for mainland students.

Of course coordinate geometry still works well for this problem, as always. Yet one must be caution about the 0 or 7 nature of coordinate geometry solutions, and should find some pure geometry clues before really attempting co-geom.

For those who are going to CMO soon, I would say it would be much better if you try to grasp the feeling of mainland contests first instead of treating it as a general mathematical olympiad. Even if the solutions look scary, sometimes the idea flow is not really *that* unapproachable — it’s just technical. After all, you have much time to execute everything out!